

# PROJECT: RETAIL SALES ANALYTICS USING SNOWFLAKE

## End-to-End Data Engineering Project

### STEP-1: BUSINESS REQUIREMENT

#### Project Overview

This project demonstrates the implementation of an automated Retail Sales Analytics pipeline using Snowflake. Multiple retail stores submit daily sales files to Head Office. The pipeline ingests these files into Snowflake, captures data changes using Streams, processes incremental records using Tasks and MERGE operations, and maintains a curated reporting table.

#### Business Objective

- Automate daily sales data processing
- Handle Inserts, Updates, and Deletes
- Maintain a reporting-ready dataset
- Reduce manual intervention
- Enable reliable business reporting

#### Architecture Flow

Retail Stores  
↓  
Daily Sales Files  
↓  
Stage  
↓  
SALES\_RAW  
↓  
STREAM  
↓  
TASK + MERGE  
↓  
SALES\_CURATED  
↓  
Reports

### STEP2 : PLATFORM SETUP

A dedicated Snowflake environment was created for the Retail Sales Analytics project to ensure logical separation from other development activities.

**Database: RETAIL\_SALES\_DB**

**Schemas: RAW & CURATED**

**Warehouse: RETAIL\_WH (XSMALL)**

#### Purpose

- Isolate project objects
- Improve organization
- Simulate enterprise project structure
- Simplify maintenance and troubleshooting

((SQL script of platform setup is attached into RETAIL\_SALES\_OBJECTS.sql))

# Warehouse created

26

27

SHOW WAREHOUSES LIKE 'RETAIL\_WH';

Results (1 minute ago)

Table

Chart

Pivot

|   | name      | state     | type     | size    |
|---|-----------|-----------|----------|---------|
| 1 | RETAIL_WH | SUSPENDED | STANDARD | X-Small |

# Database created

USE DATABASE RETAIL\_SALES\_DB;  
SELECT CURRENT\_DATABASE();

Results (just now)

Table

Chart

Pivot

| CURRENT_DATABASE() |
|--------------------|
| RETAIL_SALES_DB    |

# Schemas created

18

19

20

SHOW SCHEMAS IN DATABASE RETAIL\_SALES\_DB;

Results (1 minute ago)

Table

Chart

Pivot

|   | created_on                    | name               | is_default | is_current | database_name   | owner        |
|---|-------------------------------|--------------------|------------|------------|-----------------|--------------|
| 1 | 2026-09-14 00:48:52.217 -0700 | CURATED            | N          | N          | RETAIL_SALES_DB | ACCOUNTADMIN |
| 2 | 2026-09-14 04:53:34.248 -0700 | INFORMATION_SCHEMA | N          | N          | RETAIL_SALES_DB |              |
| 3 | 2026-09-14 00:48:40.777 -0700 | PUBLIC             | N          | Y          | RETAIL_SALES_DB | ACCOUNTADMIN |
| 4 | 2026-09-14 00:48:48.853 -0700 | RAW                | N          | N          | RETAIL_SALES_DB | ACCOUNTADMIN |

STEP 3: OBJECT CREATION

The core Snowflake objects required for the Retail Sales Analytics pipeline were created in the project environment. These objects collectively enable automated ingestion, change data capture, incremental processing, and reporting.

Objects Created

- File Format
  - Defines the structure of incoming CSV files.
- Internal Stage
  - Stores source sales files before loading into Snowflake tables.
- SALES\_RAW
  - Landing table for incoming sales data.

- **SALES\_CURATED**
  - Reporting-ready table containing the latest sales records.
- **SALES\_STREAM**
  - Tracks changes occurring in the SALES\_RAW table.
- **MERGE SQL**
  - Handles Insert, Update, Delete ingestion under Task into the CURATED.
- **SALES\_TASK**
  - Automates incremental processing from RAW to CURATED.

## MERGE & Task Processing

- SALES\_STREAM captures changes from SALES\_RAW.
- MERGE applies INSERT, UPDATE, and DELETE operations to SALES\_CURATED.
- SALES\_TASK executes the MERGE process automatically every minute.

## Pipeline Flow

## Sales File

↓

Stage

↓

SALES\_RAW

↓

SALES\_STREAM

↓

SALES\_TASK

↓

MERGE

↓

SALES\_CURATED

*((SQL script of objects creation is attached into RETAIL\_SALES\_OBJECTS.sql))*

## All Objects Screenshots

```
# File format created
```

```
14 SHOW FILE FORMATS LIKE 'SALES_CSV_FORMAT';
```

Results (just now)

Table Chart Pivot 🔍 🗒 1 row ⓘ 69ms

|   | 🕒 created_on        | 🔗 name           | 🔗 database_name | 🔗 schema_name | 🔗 type | 🔗 owner      | 🔗 comment | 🔗 format_options     |
|---|---------------------|------------------|-----------------|---------------|--------|--------------|-----------|----------------------|
| 1 | 2026-09-14 07:53:59 | SALES_CSV_FORMAT | RETAIL_SALES_DB | RAW           | CSV    | ACCOUNTADMIN |           | ("TYPE":"CSV","RECOR |

```
# Stage created
```

|                    |                                                     |
|--------------------|-----------------------------------------------------|
| 13                 |                                                     |
| 14                 | SHOW STAGES LIKE 'SALES_STAGE';                     |
| Results (just now) |                                                     |
| Table Chart Pivot  |                                                     |
|                    | created_on name database_name schema_name           |
| 1                  | 2026-09-14 07:54:02 SALES_STAGE RETAIL_SALES_DB RAW |

# Raw table created

|                    |                                                         |
|--------------------|---------------------------------------------------------|
| 13                 |                                                         |
| 14                 | SHOW TABLES LIKE 'SALES_RAW';                           |
| Results (just now) |                                                         |
| Table Chart Pivot  |                                                         |
|                    | created_on name database_name schema_name kind          |
| 1                  | 2026-09-14 07:54:28 SALES_RAW RETAIL_SALES_DB RAW TABLE |

# Stream created

|                         |                                                         |
|-------------------------|---------------------------------------------------------|
| 14                      |                                                         |
| 15                      | SHOW STREAMS LIKE 'SALES_STREAM';                       |
| Results (3 minutes ago) |                                                         |
| Table Chart Pivot       |                                                         |
|                         | created_on name database_name schema_name kind          |
| 1                       | 2026-09-14 07:54:28 SALES_RAW RETAIL_SALES_DB RAW TABLE |

# Task created

|                                                                                    |                                                                                                      |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 15                                                                                 | SHOW TASKS LIKE 'SALES_TASK';                                                                        |
| Results (just now)                                                                 |                                                                                                      |
| Table Chart Pivot                                                                  |                                                                                                      |
| <div> <div>1 row</div> <div>182ms</div> <div></div> <div></div> <div></div> </div> |                                                                                                      |
|                                                                                    | created_on name id database_name schema_name owner comment warehouse schedule                        |
| 1                                                                                  | 2026-09-14 08:02:53 SALES_TASK 01c71146-11e2-66d RETAIL_SALES_DB RAW ACCOUNTADMIN RETAIL_WH 1 MINUTE |

# Curated table created

|                    |                                                                 |
|--------------------|-----------------------------------------------------------------|
| 14                 | USE SCHEMA CURATED;                                             |
| 15                 | SHOW TABLES LIKE 'SALES_CURATED';                               |
| Results (just now) |                                                                 |
| Table Chart Pivot  |                                                                 |
|                    | created_on name database_name schema_name kind                  |
| 1                  | 2026-09-14 07:54:52 SALES_CURATED RETAIL_SALES_DB CURATED TABLE |

STEP 4 - INITIAL DATA LOAD

A sales file containing 50 transactions was received from retail stores and loaded into the Snowflake pipeline.

Objective

- Ingest source sales data into the RAW layer



# Store wise Raw table count

20  
21  
22

```
SELECT STORE_ID, COUNT(*) FROM RAW.SALES_RAW GROUP BY STORE_ID ORDER BY STORE_ID;
```

|                   |          |              |
|-------------------|----------|--------------|
| Table Chart Pivot |          | 5 rows 116ms |
| 00                | STORE_ID | # COUNT(*)   |
| 01                | S001     | 10           |
| 02                | S002     | 10           |
| 03                | S003     | 10           |
| 04                | S004     | 10           |
| 05                | S005     | 10           |

# Duplicates check

25  
26

```
--Duplicates check  
SELECT SALE_ID, COUNT(*) FROM RAW.SALES_RAW GROUP BY SALE_ID HAVING COUNT(*) > 1;
```

just now

|                   |         |
|-------------------|---------|
| Table Chart Pivot |         |
| 00                | SALE_ID |



Query produced no results

# Stream count

6  
7

```
SELECT COUNT(*) FROM RAW.SALES_STREAM;
```

results (just now)

|                   |            |
|-------------------|------------|
| Table Chart Pivot |            |
| 00                | # COUNT(*) |
| 01                | 50         |

# Stream status

29  
30  
31

```
SELECT METADATA$ACTION, COUNT(*) FROM RAW.SALES_STREAM GROUP BY METADATA$ACTION;
```

results (just now)

|                   |                  |             |
|-------------------|------------------|-------------|
| Table Chart Pivot |                  | 1 row 249ms |
| 00                | METADATA\$ACTION | # COUNT(*)  |
| 01                | INSERT           | 50          |

# Task resumed

|                   |                                  |
|-------------------|----------------------------------|
| 13                |                                  |
| 14                | ALTER TASK SALES_TASK RESUME;    |
| 15                |                                  |
| sults (just now)  |                                  |
| Table Chart Pivot |                                  |
| 10                | A status                         |
| 11                | Statement executed successfully. |

# Curated table count

|                   |                                             |
|-------------------|---------------------------------------------|
| 13                |                                             |
| 14                | SELECT COUNT(*) FROM CURATED.SALES_CURATED; |
| 15                |                                             |
| sults (just now)  |                                             |
| Table Chart Pivot |                                             |
| 10                | # COUNT(*)                                  |
| 11                | 50                                          |

# Store wise Curated table count

|                   |                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------|
| 1                 | SELECT STORE_ID, COUNT(*) FROM CURATED.SALES_CURATED GROUP BY STORE_ID ORDER BY STORE_ID; |
| 2                 |                                                                                           |
| 3                 |                                                                                           |
| 4                 |                                                                                           |
| sults (just now)  |                                                                                           |
| Table Chart Pivot |                                                                                           |
| 10                | A STORE_ID : # COUNT(*)                                                                   |
| 11                | S001 10                                                                                   |
| 12                | S002 10                                                                                   |
| 13                | S003 10                                                                                   |
| 14                | S004 10                                                                                   |
| 15                | S005 10                                                                                   |

# Raw & Curated counts comparison

|                   |                                                                |
|-------------------|----------------------------------------------------------------|
| 7                 | --Compare Raw & Curated after Task processing                  |
| 8                 | SELECT                                                         |
| 9                 | (SELECT COUNT(*) FROM RAW.SALES_RAW) AS RAW_COUNT,             |
| 10                | (SELECT COUNT(*) FROM CURATED.SALES_CURATED) AS CURATED_COUNT; |
| 11                |                                                                |
| sults (just now)  |                                                                |
| Table Chart Pivot |                                                                |
| 10                | # RAW_COUNT # CURATED_COUNT                                    |
| 11                | 50 50                                                          |

# Stream status post Task execution

|                   |                                                        |
|-------------------|--------------------------------------------------------|
| 16                | SELECT COUNT(*) AS RECORD_COUNT FROM RAW.SALES_STREAM; |
| 17                |                                                        |
| 18                |                                                        |
| sults (just now)  |                                                        |
| Table Chart Pivot |                                                        |
| 10                | # RECORD_COUNT                                         |
| 11                | 0                                                      |

## STEP-5: DUPLICATE LOAD TEST

A duplicate load scenario was executed by reprocessing an already ingested sales file.

### Objective

- This is to validate Snowflake file load behavior.
- And understand duplicate data impact on the pipeline.
- Also to verify the MERGE handling in the curated layer.

### Observations

- The source file was reloaded using FORCE=TRUE.
- Duplicate records were inserted into SALES\_RAW.
- Changes were captured by SALES\_STREAM.
- SALES\_TASK processed the changes automatically.
- SALES\_CURATED retained the correct business view through MERGE processing.

### Outcome

- RAW layer contained duplicate source records.
- Stream captured the duplicate load activity.
- Curated layer maintained expected record counts and data consistency.

*((SQL scripts of COPY INTO & Validation queries are attached into RETAIL\_SALES\_VALIDATION\_QUERIES.sql))*

# Attempting duplication scenario – Copying same day1 sales data to the Raw table forcefully

|                    |                                                             |
|--------------------|-------------------------------------------------------------|
| 14                 |                                                             |
| 15                 | ✓                                                           |
| 16                 | <code>COPY INTO RAW.SALES_RAW</code>                        |
| 17                 | <code>FROM @RAW.SALES_STAGE/day1_sales.csv</code>           |
| 18                 | <code>FILE_FORMAT = (FORMAT_NAME = SALES_CSV_FORMAT)</code> |
| 19                 | <code>FORCE = TRUE;</code>                                  |
| results (just now) |                                                             |
| Table Chart Pivot  |                                                             |
| #                  | file status # rows_parsed # rows_loaded                     |
| 1                  | sales_stage/day1_sales LOADED 50 50                         |

# Duplicates loaded forcefully

|                    |                                                        |
|--------------------|--------------------------------------------------------|
| 22                 |                                                        |
| 23                 | ✓                                                      |
| 24                 | <code>SELECT COUNT(*) FROM RAW.SALES_RAW; --100</code> |
| results (just now) |                                                        |
| Table Chart Pivot  |                                                        |
| #                  | # COUNT(*)                                             |
| 1                  | 100                                                    |

# Due to Matched condition in Merge in Task, few fields are updated

# for the matching sales ids

|                    |                                                      |
|--------------------|------------------------------------------------------|
| 31                 |                                                      |
| 32                 | SELECT COUNT(*) FROM CURATED.SALES_CURATED; --old 50 |
| 33                 |                                                      |
| 34                 |                                                      |
| Results (just now) |                                                      |
| Table Chart Pivot  |                                                      |
| 00                 | # COUNT(*)                                           |
| 1                  | 50                                                   |

STEP-6: UPDATE & DELETE TEST

A change file containing new sales, corrected sales, and cancelled sales transactions was processed through the pipeline.

Changes Applied

- New sales records inserted
- Existing sales records updated
- Cancelled sales records removed

Validation

- Changes captured by SALES\_STREAM
- SALES\_TASK executed automatically
- MERGE synchronized SALES\_CURATED
- Final curated dataset reflected the latest business state

((SQL scripts of COPY INTO & Validation queries are attached into RETAIL\_SAELS\_VALIDATION\_QUERIES.sql))

# Loading day2 file for corrections - update & delete – 10 rows

|                    |                                                 |
|--------------------|-------------------------------------------------|
| 13                 |                                                 |
| 14                 | COPY INTO RAW.SALES_RAW                         |
| 15                 | FROM @RAW.SALES_STAGE/day2_sales.csv            |
| 16                 | FILE_FORMAT = (FORMAT_NAME = SALES_CSV_FORMAT); |
| 17                 |                                                 |
| 18                 |                                                 |
| Results (just now) |                                                 |
| Table Chart Pivot  |                                                 |
| 00                 | file status # rows_parsed # rows_loaded         |
| 1                  | sales_stage/day2_sales LOADED 10 10             |

# New count of Raw table post day2 upload

|                    |                                     |
|--------------------|-------------------------------------|
| 1                  |                                     |
| 2                  | SELECT COUNT(*) FROM RAW.SALES_RAW; |
| 3                  |                                     |
| Results (just now) |                                     |
| Table Chart Pivot  |                                     |
| 00                 | # COUNT(*)                          |
|                    | 110                                 |

# CDC

|                   |                                        |
|-------------------|----------------------------------------|
| 4                 |                                        |
| 5                 |                                        |
| 6                 | SELECT COUNT(*) FROM RAW.SALES_STREAM; |
| sults (just now)  |                                        |
| Table Chart Pivot |                                        |
| 00                | # COUNT(*)                             |
| 1                 | 10                                     |

# Curated count remain same even after day2 processed

|                   |                                             |
|-------------------|---------------------------------------------|
| 9                 |                                             |
| 0                 | SELECT COUNT(*) FROM CURATED.SALES_CURATED; |
| 1                 |                                             |
| sults (just now)  |                                             |
| Table Chart Pivot |                                             |
| 00                | # COUNT(*)                                  |
| 1                 | 50                                          |

# Store wise count in Curated post day2 upload

|                   |                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------|
| 33                |                                                                                           |
| 34                | SELECT STORE_ID, COUNT(*) FROM CURATED.SALES_CURATED GROUP BY STORE_ID ORDER BY STORE_ID; |
| 35                |                                                                                           |
| sults (just now)  |                                                                                           |
| Table Chart Pivot |                                                                                           |
| 5 rows 120ms      |                                                                                           |
| 00                | STORE_ID # COUNT(*)                                                                       |
| 1                 | S001 11                                                                                   |
| 2                 | S002 11                                                                                   |
| 3                 | S003 11                                                                                   |
| 4                 | S004 10                                                                                   |
| 5                 | S005 7                                                                                    |

### STEP-7: REPORTING QUERIES

The curated sales data was queried to generate business-level reporting outputs. These queries demonstrate that the processed data is ready for downstream analytics and reporting.

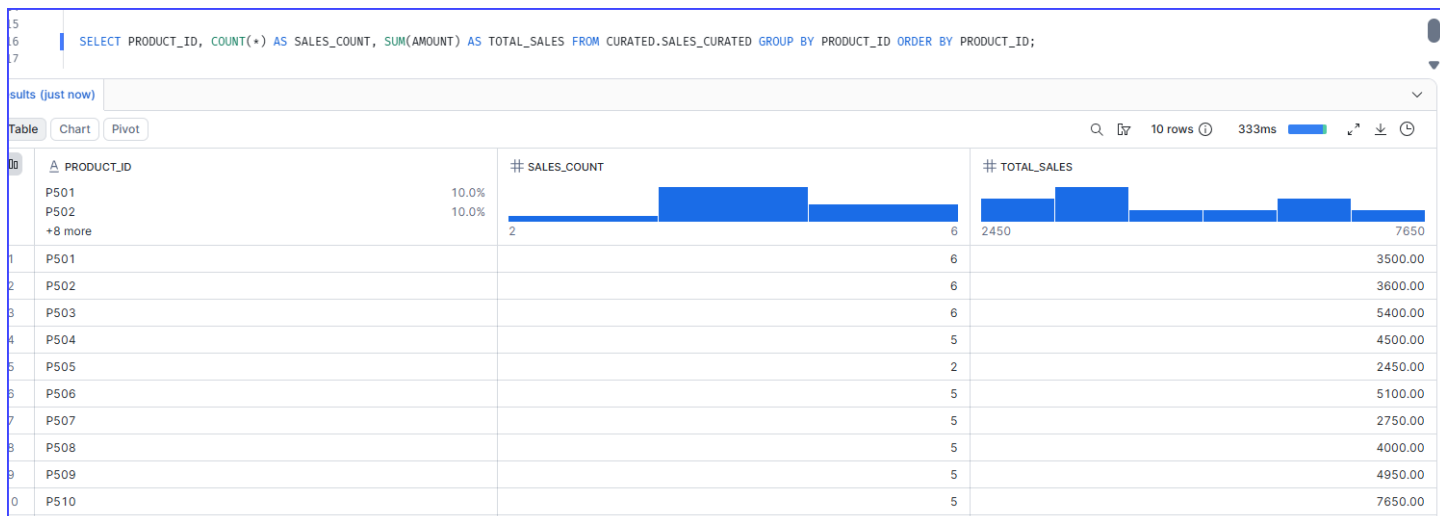
#### Reports Created

- Store-wise Sales Summary
- Product-wise Sales Summary
- Daily Sales Summary

# Store-wise Sales

|                   |                                                                                                                                      |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 15                |                                                                                                                                      |
| 16                | SELECT STORE_ID, COUNT(*) AS SALES_COUNT, SUM(AMOUNT) AS TOTAL_SALES FROM CURATED.SALES_CURATED GROUP BY STORE_ID ORDER BY STORE_ID; |
| 17                |                                                                                                                                      |
| sults (just now)  |                                                                                                                                      |
| Table Chart Pivot |                                                                                                                                      |
| 5 rows 201ms      |                                                                                                                                      |
| 00                | STORE_ID # SALES_COUNT # TOTAL_SALES                                                                                                 |
| 1                 | S001 11 8600.00                                                                                                                      |
| 2                 | S002 11 6350.00                                                                                                                      |
| 3                 | S003 11 9400.00                                                                                                                      |
| 4                 | S004 10 9450.00                                                                                                                      |
| 5                 | S005 7 10100.00                                                                                                                      |

# Product-wise Sales



## # Daily Sales Summary

```

28
29 | SELECT SALE_DATE, COUNT(*) AS SALES_COUNT, SUM(QUANTITY) AS TOTAL_QUANTITY, SUM(AMOUNT) AS TOTAL_SALES FROM CURATED.SALES_CURATED GROUP BY SALE_DATE ORDER BY SALE_DATE;
30

```

results (just now)

Table Chart Pivot

2 rows 442ms

| SALE_DATE  | SALES_COUNT | TOTAL_QUANTITY | TOTAL_SALES |
|------------|-------------|----------------|-------------|
| 2026-09-01 | 43          | 107            | 34300.00    |
| 2026-09-02 | 7           | 25             | 9600.00     |

## STEP-8: PROJECT SUMMARY & KEY LEARNINGS

### Project Summary

The Retail Sales Analytics project implemented an end-to-end incremental data pipeline in Snowflake for processing daily sales data from multiple retail stores.

### Key Snowflake Features Used

- Internal Stage and File Format
- RAW and CURATED data layers
- Streams for change tracking
- Tasks for automated processing
- MERGE for INSERT, UPDATE & DELETE handling
- Duplicate file-load handling
- Reporting-ready queries

### Key Learnings

- Built a complete pipeline, from File → Stage → RAW → Stream → Task/MERGE → CURATED pipeline.
- Handled initial, duplicate, updated and cancelled sales records.
- Validated data at RAW, Stream and CURATED layers.
- Understood Snowflake's file-load history and controlled duplicate reprocessing using FORCE=TRUE.
- Maintained the latest business-ready data in the CURATED layer.

### Future Enhancements

- Snowpipe-based automated ingestion

- External cloud storage integration
- JSON/Parquet source formats
- Larger datasets for performance testing
- BI/dashboard integration

### **Outcome**

The project successfully demonstrated a practical **Snowflake incremental data pipeline** from source-file ingestion through reporting-ready data.